Ip addressing and subnetting

Q-1 List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Ans: IP Address Classes and Real-World Examples

IP Address Class	IP Address Range	Purpose	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Used for very large networks.	A large organization with many connected devices.
Class B	128.0.0.0 – 191.255.255.255	Used for medium to large networks.	A university campus network.
Class C	192.0.0.0 – 223.255.255.255	Used for smaller networks.	A home router's private LAN, such as 192.168.1.10.
Class D	224.0.0.0 – 239.255.255.255	Used for multicast communication.	A live video streaming service sending data to multiple receivers through multicast.
Class E	240.0.0.0 – 255.255.255.255	Reserved for experimental or future use.	Network researchers testing experimental IP configurations.

Q-2 Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.

Ans: Given Information

- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0

Step 1: Calculate the Network Address

The subnet mask 255.255.255.0 means the first three octets represent the network, and the last octet represents the host.

IP Address: 192.168.10.45

Subnet Mask: 255.255.255.0

Set the host portion to 0.

Network Address = 192.168.10.0

Step 2: Calculate the Broadcast Address

To find the broadcast address, set all host bits to 1. With this subnet mask, the last octet becomes 255.

Broadcast Address = 192.168.10.255

Step 3: Calculate the Valid Host Range

- First valid host = Network address + 1
- Last valid host = Broadcast address − 1

Therefore:

- First valid host: 192.168.10.1
- Last valid host: 192.168.10.254

The number of usable host addresses is:

\[2^8 - 2 = 256 - 2 = 254 \]

We subtract 2 because the network address and broadcast address cannot be assigned to regular hosts.

Final Answer

Item	**Result**
IP Address	192.168.10.45
Subnet Mask	255.255.255.0 (/24)
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
Valid Host Range	192.168.10.1 – 192.168.10.254
Total Usable Host Addresses	254

Q-3 Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Ans: Subnetting Plan for a Food Delivery Startup Like Zomato

The startup has the network 10.0.0.0/24 and needs four separate subnets for Delivery Partners, Restaurants, Customers, and Admins.

To create four equal subnets, we borrow 2 bits from the host portion:

- Original subnet mask: 255.255.255.0 (/24)

- New subnet mask: 255.255.255.192 (/26)
- Number of subnets: \(2^2 = 4\)
- Usable hosts per subnet: \(2^6 - 2 = 62\)

Subnet Allocation Table

Group	Network Address	Subnet Mask	Valid Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Explanation

1. The original /24 network is divided into four equal /26 subnets.
2. Each subnet contains 64 total IP addresses, including the network and broadcast addresses.
3. Each subnet supports 62 usable host addresses.
4. Separating the groups makes it easier to manage network traffic and apply security rules. For example, the Admin subnet can be restricted to authorized staff.

Q-4 You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.

Ans: Subnetting Calculation for a College Wi-Fi Network

Given:

- IP Network: 172.16.0.0/16
- Required subnets: At least 20
- Goal: Calculate the new subnet mask and usable hosts per subnet.

Step 1: Calculate the Bits Required for Subnets

We use the formula:

\[2^n \geq \text{Required Subnets} \]

Here, \(n\) is the number of bits borrowed from the host portion.

- \(2^4 = 16\) subnets — not enough.
- \(2^5 = 32\) subnets — enough.

Therefore, we must borrow 5 bits.

Step 2: Calculate the New Subnet Mask

The original prefix is /16.

\[16 + 5 = 21 \]

The new prefix is /21.

The subnet mask for /21 is:

255.255.248.0

Step 3: Calculate Hosts per Subnet

The original network has 16 host bits. After borrowing 5 bits:

\[16 - 5 = 11 \text{ host bits} \]

Total addresses per subnet:

\[2^{11} = 2048 \]

Usable host addresses:

\[2048 - 2 = 2046 \]

We subtract 2 for the network address and broadcast address.

Final Answers

Item	**Answer**
Original Network	172.16.0.0/16
Required Subnets	At least 20
Bits Borrowed	5 bits
Total Subnets Created	32
New Subnet Mask	255.255.248.0
New CIDR Prefix	/21
Total Addresses per Subnet	2048
Usable Hosts per Subnet	2046

Q-5 A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Ans: Explanation of the Invalid IP Address

The IP address 192.168.1.300 is invalid because each octet in an IPv4 address must be between 0 and 255. Here, the last octet 300 exceeds the maximum allowed value.

A valid IP address in the same subnet could be:

- 192.168.1.10
- 192.168.1.100
- 192.168.1.254

These examples are valid host addresses assuming the subnet mask is 255.255.255.0 (/24) and the addresses are not already in use.